AI-Driven Crop Yield Prediction System

January Progress Report

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1 Draft Contents Page for Final Dissertation

2 Technical Chapter: Methodology

This chapter presents the methodology for developing regional crop yield prediction models using weather data and machine learning.

2.1 Feature Engineering

2.1.1 Overview

Raw monthly weather data (rainfall, temperature, frost, sunshine) were transformed into 47 seasonal features aligned with crop growth stages. This is crucial because crops respond to weather during specific phenological periods, not uniformly throughout the year.

2.1.2 Seasonal Framework

Weather variables were aggregated into four seasons matching crop growth cycles:

- **Winter** (Dec-Feb): Coldness, winter hardening
- **Spring** (Mar-May): Growth, tillering, emergence
- **Summer** (Jun-Aug): Flowering, grain filling, harvest
- **Autumn** (Sep-Nov): Planting, establishment

2.1.3 Features (47 total)

- **Temperature (23):** Seasonal max/min, critical period temps, extremes, growing degree days
- **Rainfall (14):** Seasonal totals, critical periods, optimal deviation, non-linear terms
- **Sunshine (7):** Seasonal totals, critical period sunshine, efficiency ratios
- **Frost (4):** Seasonal frost days, late spring frost risk
- **Interactions (7):** Temperature $\times$ rainfall, binary extreme indicators

2.2 Modeling Approach

2.2.1 Train-Test Split

Temporal splitting was used to simulate real-world forecasting:

- **Training:** 2004-2019 (64 observations, 76%)
- **Testing:** 2020-2024 (20 observations, 24%)

2.2.2 Model Selection

Two approaches were compared:

1. Ridge Regression (linear baseline) with L2 regularization

2. Random Forest (non-linear) chosen for its ability to capture interactions, robustness to outliers, and interpretable feature importance. Also, proved to be the best model for crop prediction according to the research papers on this topic.

2.2.3 Evaluation Metrics

- **R^2 (Coefficient of Determination):** Primary metric for variance explained
- **RMSE (Root Mean Squared Error):** Interpretable error in t/ha

2.3 Crop-Specific Modeling

2.3.1 Feature Selection Process

For each crop: (1) correlation testing with yield, (2) phenological alignment, (3) multi-collinearity removal, (4) test different combinations, (5) select based on test R^2.

2.3.2 Spring vs. Winter Barley Separation

An important discovery was finding out that spring and winter barley must be modeled separately due to different planting times, growth cycles, and a 1.44 t/ha (27%) yield difference. The aggregated "Barley" model failed (R^2=-0.16), while separated models achieved R^2=0.45 for winter barley.

2.3.3 Final Model Features

2.4 Extreme Year Validation

To test whether models capture genuine weather-yield relationships, validation was performed on two historically documented extreme years:

- **2010:** Coldest UK winter since 1978-79 (Winter frost: +3.5 standard deviations)
- **2012:** Wettest English summer on record (Summer rain: +1.7 SD, sunshine: -1.8 SD)

Models were trained excluding these years, then tested on them. Results showed strong predictive performance:

The models performed well on the 2010 cold winter, with all predictions within 0.30 t/ha of actual yields. For the 2012 wet summer, barley and OSR predictions remained accurate. However, the wheat model underestimated the severity of the 2012 yield crash, highlighting a limitation when conditions exceed the training data range.

2.5 Progress, Problems, and Solutions

2.5.1 Progress Made

- Expanded from national to regional data (21 → 84 observations per crop)
- Engineered 47 biologically-meaningful weather features
- Implemented and compared Baseline and Random Forest models
- Discovered need to separate spring/winter barley
- Completed extreme year validation achieving 88% direction accuracy

2.5.2 Problems and Solutions

2.5.3 Current Results

Key findings:

- Winter sunshine is a strong predictor for wheat yield (r=0.47)
- Spring/winter barley must be modeled separately (1.44 t/ha difference)
- OSR yields are pest-dominated rather than weather-dominated
- Hybrid model achieves 88% direction accuracy on extreme years

3 Revised Work Plan (January - April 2025)

3.1 January 2025 - Model Completion

- Finalize all crop-specific Random Forest models
- Document model performance and feature importance
- Compile results for all five crops (wheat, winter barley, spring barley, oats, OSR)

Deliverable: January Deliverable

3.2 February 2025 - Interface Development & Results

- Develop web-based prediction interface
- Integrate trained models for real-time yield predictions
- Implement basic risk scenario calculator to help farmers
- Draft Results chapter with visualizations

Deliverable: Working prediction interface prototype

3.3 March 2025 - Analysis & Writing

- Complete Results and Discussion chapters
- Write Literature Review
- Write Introduction (background, research questions, objectives)
- Polish prediction interface based on testing

Deliverable: First complete dissertation draft

3.4 April 2025 - Final Revisions

- Write conclusions and future work
- Final formatting, figures, tables, references

- Proofreading and quality check
- Prepare appendices (code, additional tables)

Final Submission: April, 2025